Introduction to Quantum Physics and Structure of Matter

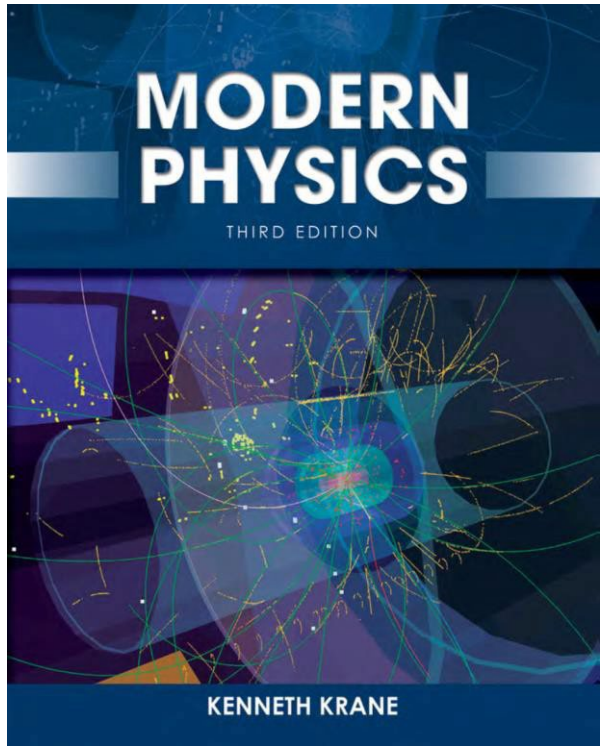
Lecture 1

October 24, 2026

EINDHOVEN UNIVERSITY OF TECHNOLOGY

Course Contents

- Physics necessary for understanding radiation phenomena
- Course necessary for certificate “Fysica in de Kliniek”
- Course structure:
  - Lectures (Wednesdays, 13:30 – 15:15)
  - Guided Self-study (Wednesdays, 15:30 – 17:30)
- Book:
  - Kenneth Krane, Modern Physics (3rd edition)



Course Contents

Lecture	Date	Topics	Material
1	02-09-2026	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	09-09-2026	•Compton scattering •De Broglie's hypothesis •Experimental evidence for De Broglie waves •Uncertainty relationships for classical particles •Heisenberg uncertainty relationships	Book: Ch. 3.4, 4.1-4.4
3	16-09-2026	•Behaviour of a wave at a boundary •Confining a particle •The Schrödinger equation •Applications of the Schrödinger equation	Book: Ch. 5.1-5.4
4	23-09-2026	•Basic properties of atoms •Scattering experiments and the Thomson model •The Rutherford nuclear atom •Line spectra •The Bohr model •The Franck-Hertz experiment •Deficiencies of the Bohr model	Book: Ch. 6.1-6.6, 6.8

Course Contents

Lecture	Date	Topics	Material
5	39-09-2026	•A one-dimensional atom •Angular momentum in the hydrogen atom •The hydrogen atom wave functions •Radial probability densities •Angular probability densities	Book: Ch. 7.1-7.7
6	07-10-2026	•The Pauli exclusion principle •Electron states in many-electron atoms •Outer electrons: screening and optical transitions •Inner electrons: absorption edges and X-rays	Book: Ch. 8.1-8.3, 8.5
7	14-10-2026	•Nuclear constituents •Nuclear sizes and shapes •Alpha decay •Beta decay •Gamma decay and nuclear excited states	Book: Ch. 12.1-12.3 12.6-12.9
8	t.b.d.	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30			

Physics at the end of 19th century

Isaac Newton
1643-1727

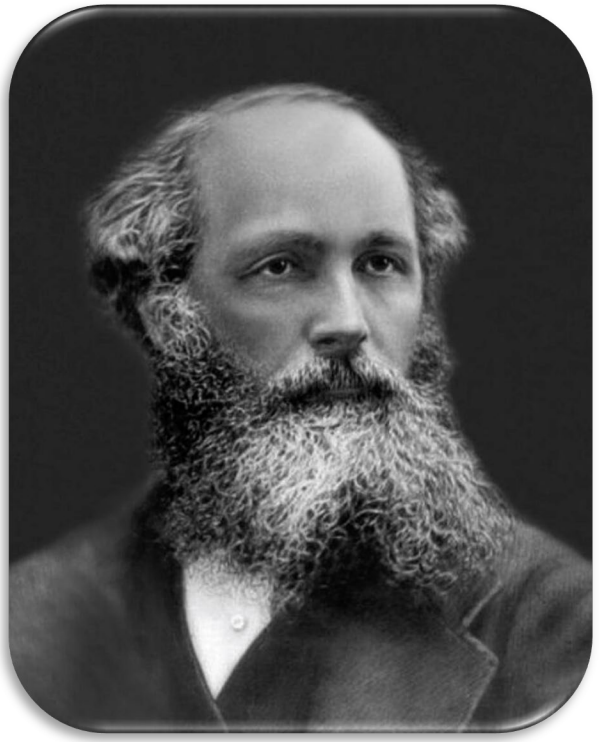


Mechanics

$$F = ma$$

$$F_{grav} = G\frac{mM}{r^2}$$

James Clerk Maxwell
1831 -1879

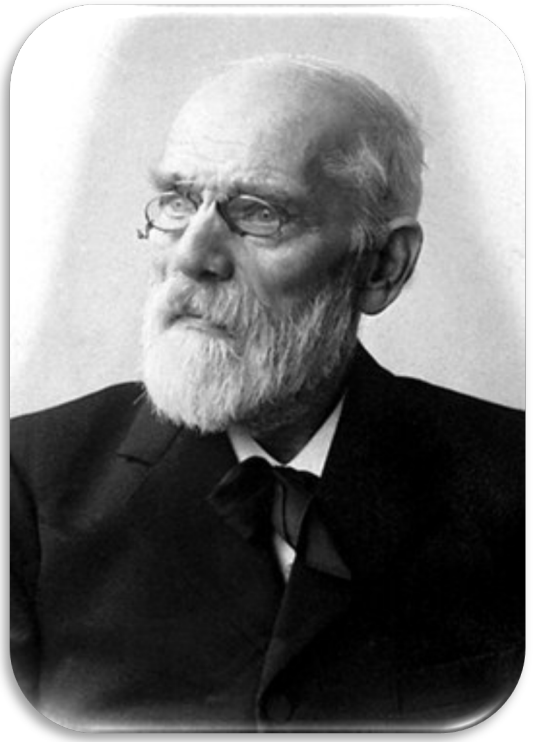


Electromagnetism

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0} \qquad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\nabla \cdot \boldsymbol{B} = 0 \qquad \nabla \times \boldsymbol{B} = \mu_0\left(\boldsymbol{J} + \epsilon_0\frac{\partial \boldsymbol{E}}{\partial t}\right)$$

Johannes Diderik van der Waals
1837 - 1923



Thermodynamics

$$\left(p + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

TU/e

Physics at the end of 19th century

- Everything is solved!

- Except…
  - Heat radiation
  - Existence of an “ether”
  - Relationship between electrodynamics & mechanics
  - Nature of radioactivity
  - Nature of light and matter

Theory of Relativity

- A “theory of relativity” is a way for observers in different frames of reference to compare the results of their observations

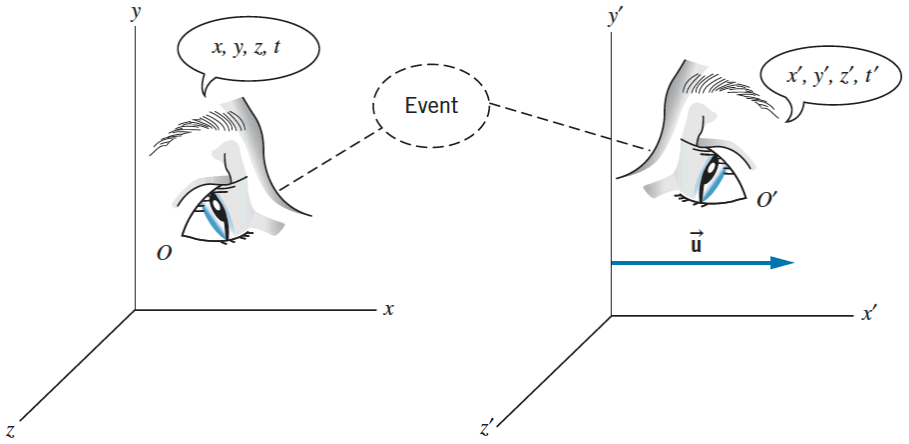


FIGURE 2.1 Two observers O and O' observe the same event. O' moves relative to O with a constant velocity $\vec{\mathbf{u}}$.

- **Inertial frame of reference:** a frame where Newton’s Laws of motion hold



Images from Unsplash

TU/e

Galilean Coordinate Transformations

$$x' = x - ut \qquad y' = y \qquad z' = z$$

$$v_x' = v_x - u \qquad v_y' = v_y \qquad v_z' = v_z$$

$$a_x' = a_x \qquad a_y' = a_y \qquad a_z' = a_z$$

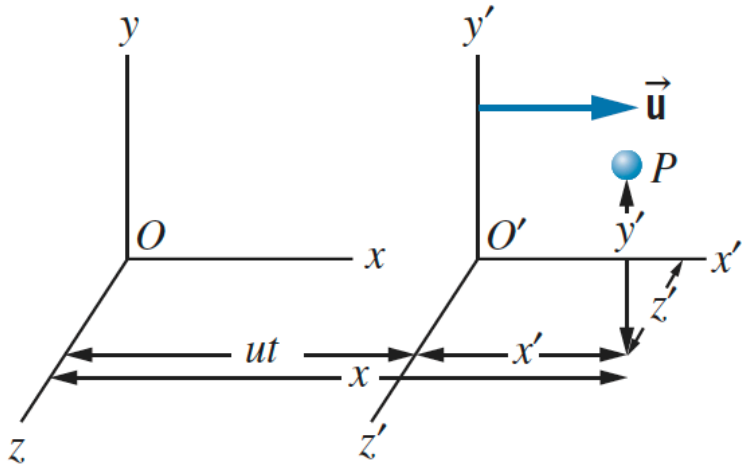
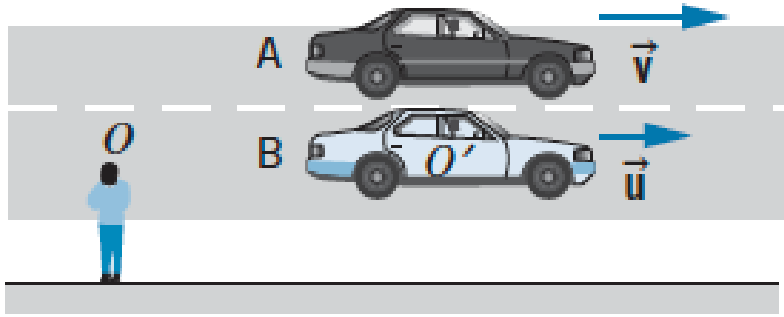


FIGURE 2.2 An object or event at point P is at coordinates x', y', z' with respect to O'. The x coordinate measured by O is $x = x' + ut$. The y and z coordinates in O are the same as those in O'.

Galilean Coordinate Transformations – Exercise

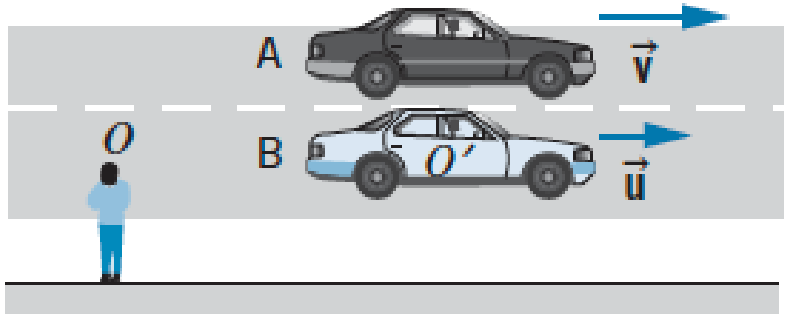
Two cars are traveling at constant speed along a road in the same direction. Car A moves at 60 km/h and car B moves at 40 km/h, each measured relative to an observer on the ground. What is the speed of car A relative to car B?



Galilean Coordinate Transformations – Exercise

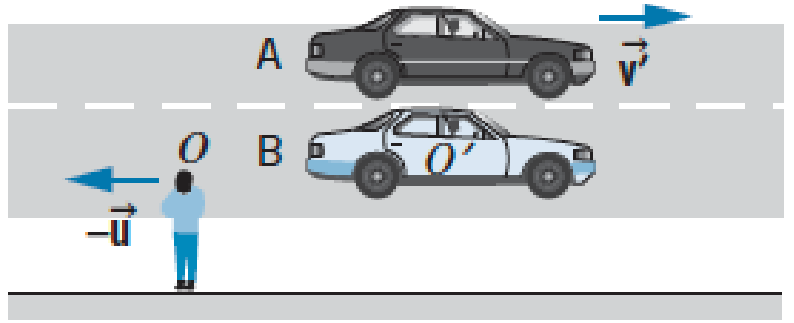
Answer:

$O: \vec{v} = 60$ km/h, $\vec{u} = 40$ km/h



(a)

$O': v_x' = v_x - u = 60$ km/h – 40 km/h
= 20 km/h

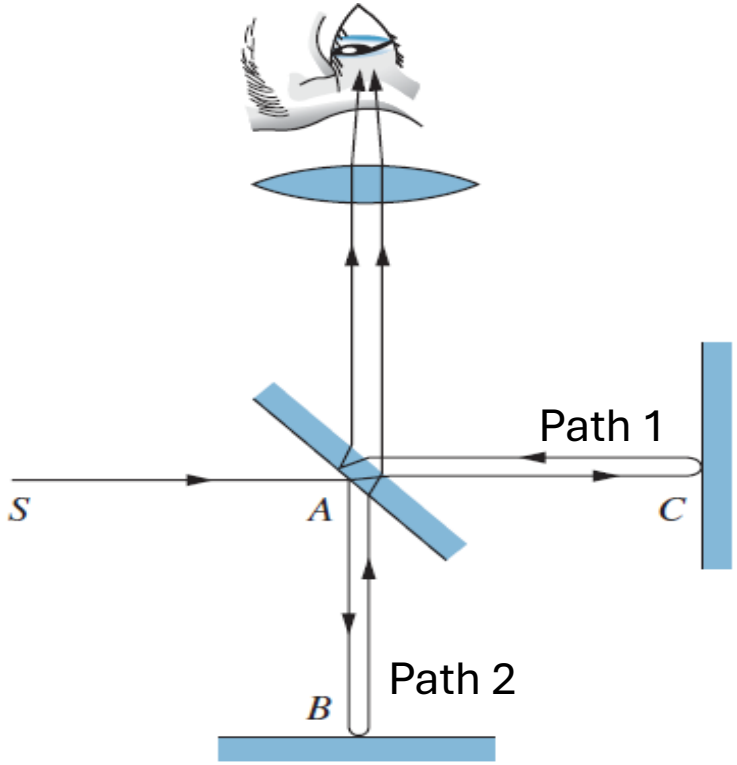


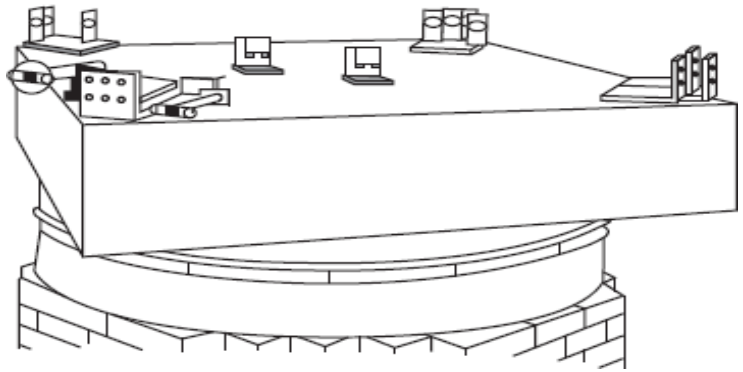
(b)

Michelson-Morley Experiment

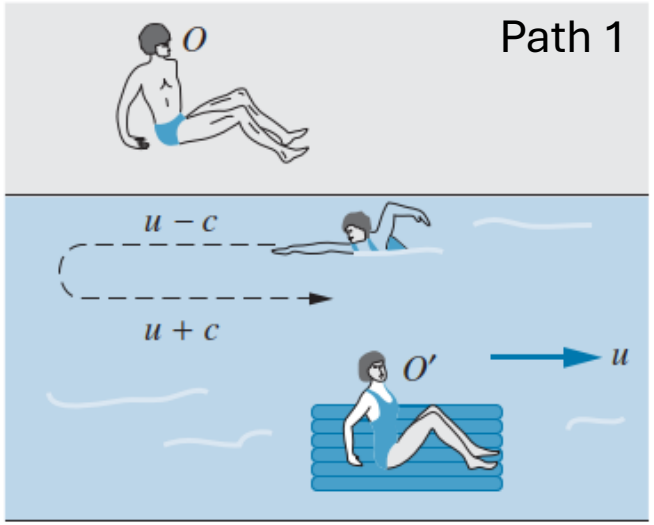
- Physicists: “Light should move through a massless medium, the *ether,* and it exists solely for the propagation of light waves”

- Measure the time difference for the two paths
  (by use of interference)





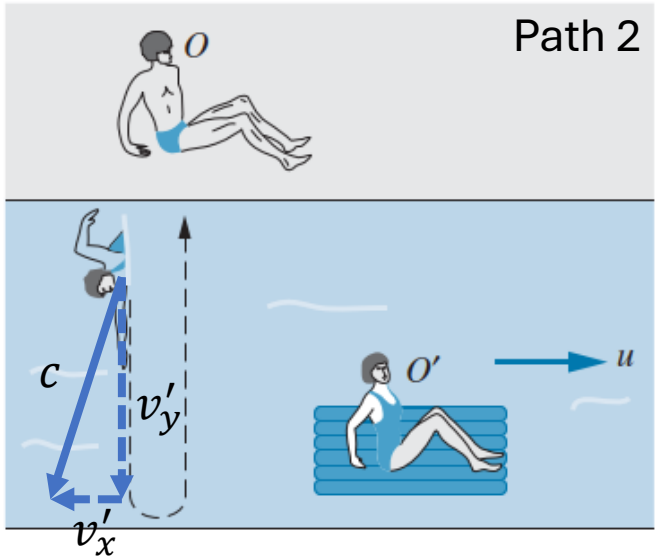
Michelson-Morley Experiment



Assuming $c > u$

Galilean transform:

$$v_x' = v_x - u$$



A moving ether should give different measurements of time!

Downstream:

$v_x = v_x' + u = u - c \qquad |v_x| = c - u$

$$t_{up} = L/(c-u)$$

Upstream:

$v_x' = c \qquad v_x = u + c$

$$t_{down} = L/(c+u)$$

$$t = t_{up} + t_{down} = \frac{2L}{c}\frac{1}{1-u^2/c^2}$$

Swimmer needs to counter the effect of the current

$v_x' = -u$ so that $v_x = 0$

$$\sqrt{{v_x'}^2 + {v_y'}^2} = c$$

$$v_y' = \sqrt{c^2 - {v_x'}^2} = \sqrt{c^2 - u^2}$$

$$t = 2t_{across} = \frac{2L}{\sqrt{c^2-u^2}} = \frac{2L}{c}\frac{1}{\sqrt{1-u^2/c^2}}$$

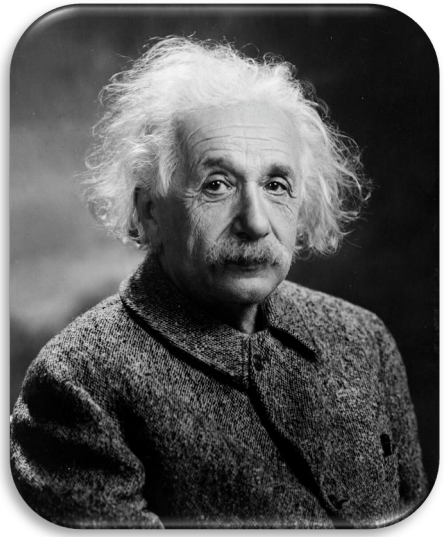
Michelson-Morley Experiment

Albert Michelson
1852 - 1931



- No observable changes in the measurements
- Conclusion: no motion of the Earth through the ether

Albert Einstein
1879 - 1955



Special Theory of Relativity

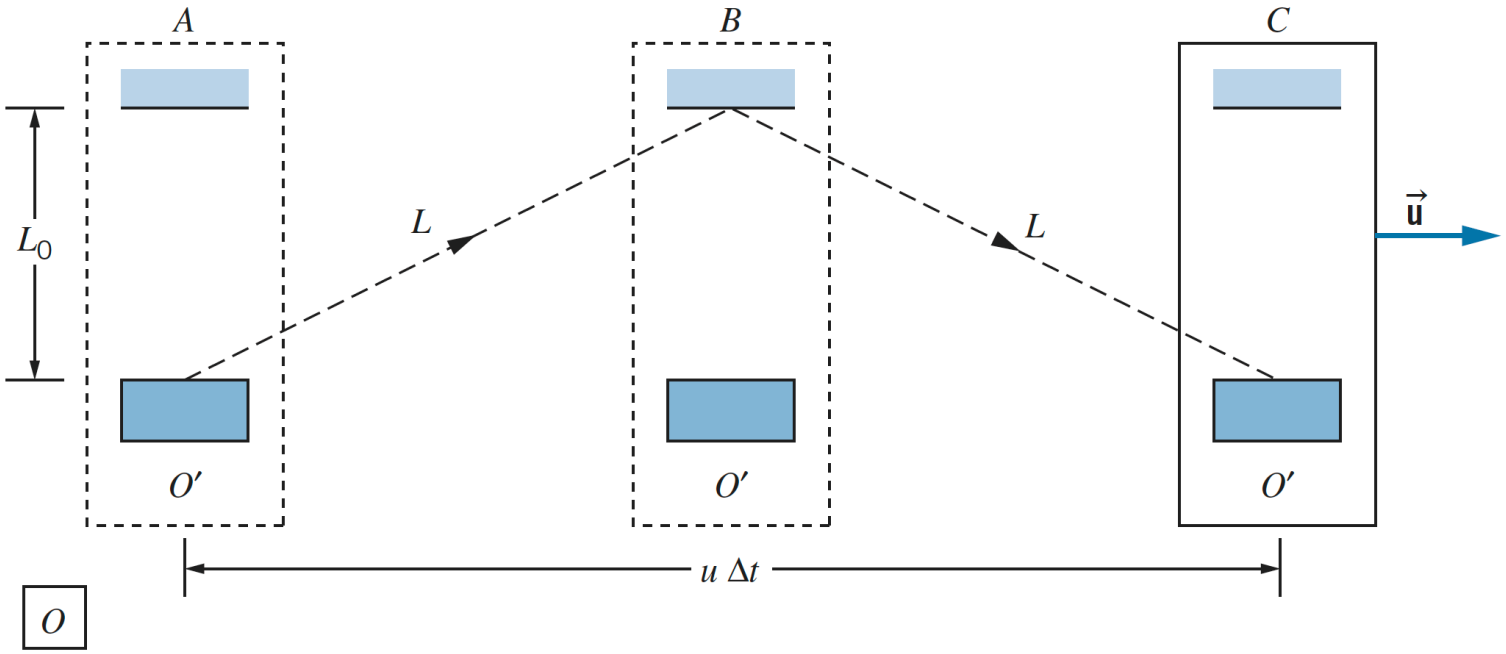
Einstein's Postulates:

1. "The laws of physics are the same in all inertial reference frames"

2. " The speed of light in free space has the same value c in all inertial reference frames"

Relativity of Time

- Consider a ‘clock’ as shown in the figure: $\Delta t_0 = \frac{2L_0}{c}$
- Now, let’s move the clock:



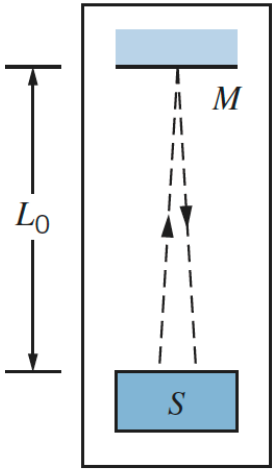
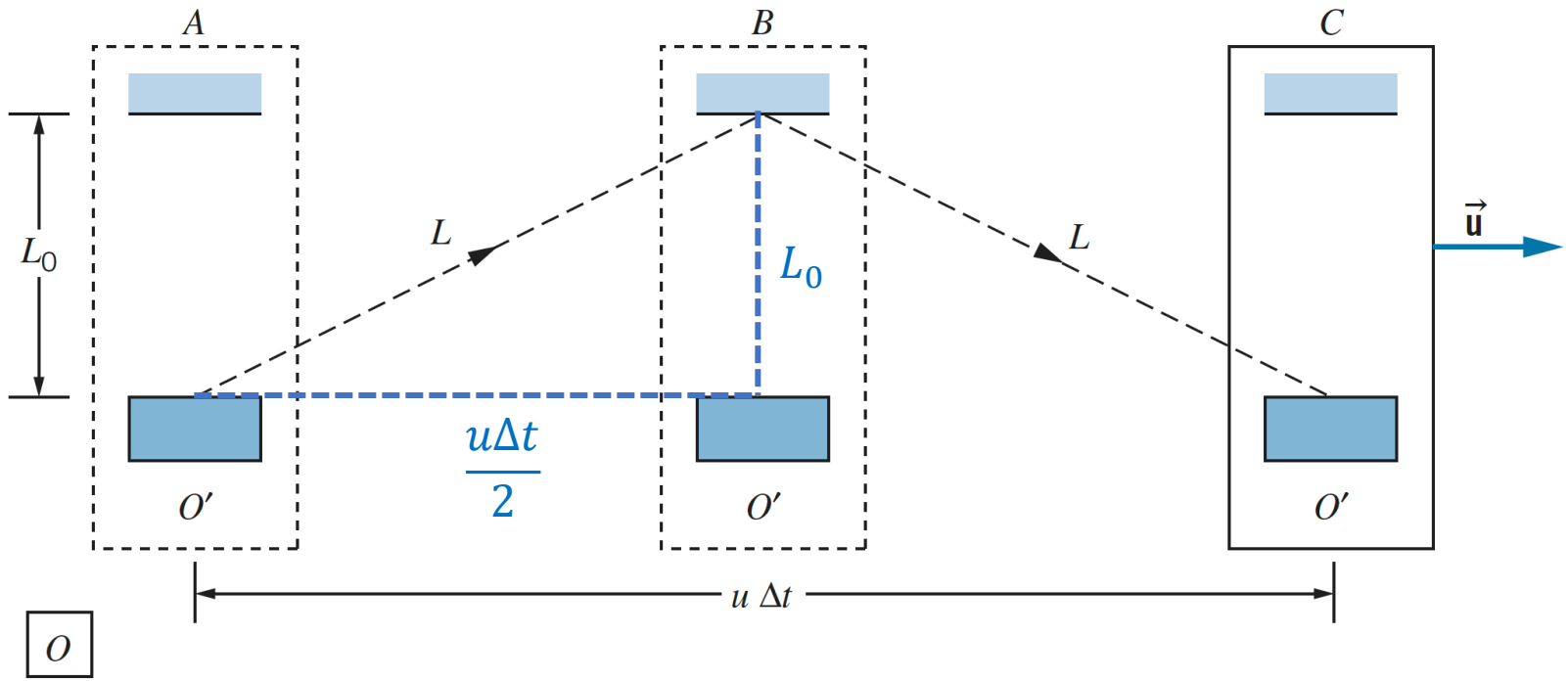


FIGURE 2.8 The clock ticks at intervals Δt_0 determined by the time for a light flash to travel the distance $2L_0$ from the light source S to the mirror M and back to the source where it is detected. (We assume the emission and detection occur at the same location, so the beam travels perpendicular to the mirror).

Relativity of Time



$$\left.\begin{aligned} O: \Delta t &= \frac{2L}{c} = \frac{2\sqrt{L_0^2 + \left(\frac{u\Delta t}{2}\right)^2}}{c} \\ O': \Delta t_0 &= \frac{2L_0}{c} \end{aligned}\right\} \Rightarrow \Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$$

Time Dilation

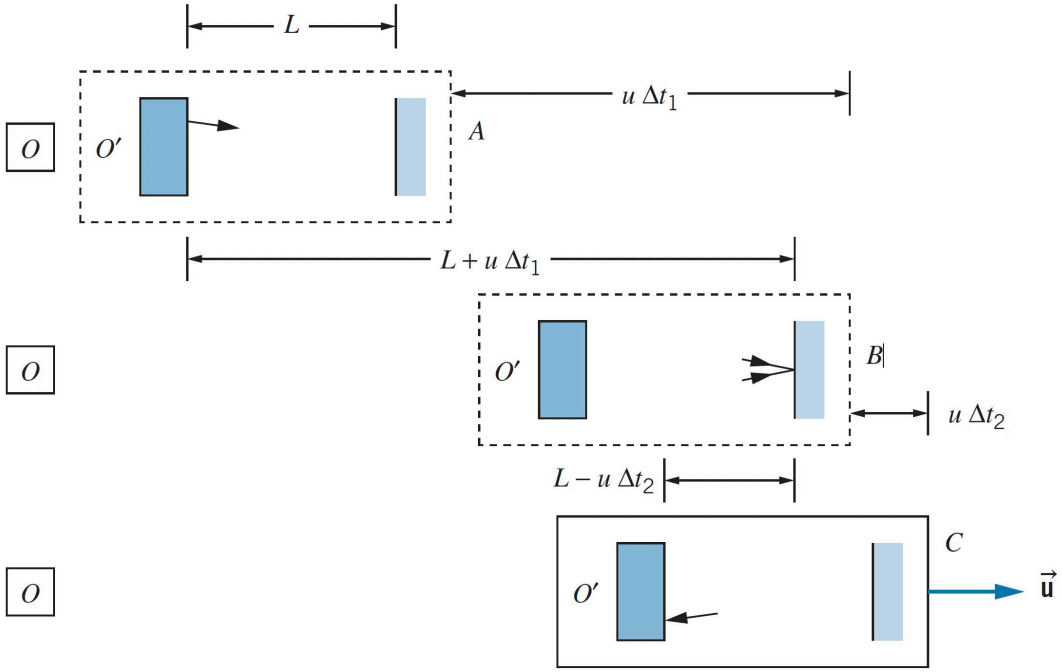
$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$$

- Δt_0 is called the “proper time”

- $\Delta t > \Delta t_0$ no matter what the direction or magnitude of $\vec{u}$ is.

Relativity of Length

We can do the same if we turn the clock sideways



$$c\Delta t_1 = L + u\Delta t_1 \quad \text{and} \quad c\Delta t_2 = L - u\Delta t_2$$

$$\Delta t = \Delta t_1 + \Delta t_2 = \frac{L}{c-u} + \frac{L}{c+u} = \frac{2L}{c}\frac{1}{1-u^2/c^2}$$

$$\text{Using: } \Delta t = \frac{\Delta t_0}{\sqrt{1-u^2/c^2}} = \frac{2L_0}{c}\frac{1}{\sqrt{1-u^2/c^2}}$$

$$L = L_0\sqrt{1-u^2/c^2}$$

Length Contraction

$$L = L_0\sqrt{1 - u^2/c^2}$$

- L_0 is called the “proper length”

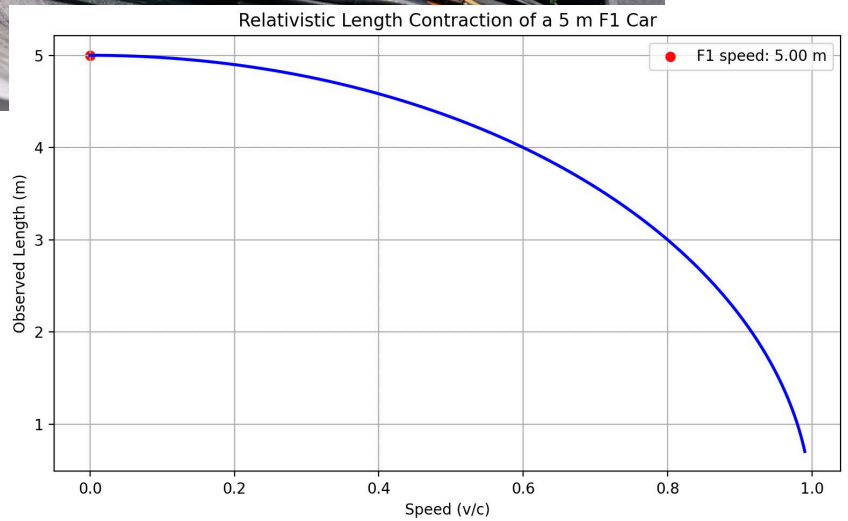
- All observers relative to whom O' is in motion measure a shorter length, but only along the direction of motion

- Common use: $\gamma = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}} \Rightarrow L = \frac{L_0}{\gamma}$ and $\Delta t = \gamma \Delta t_0$

Effects of Time Dilation & Length Contraction – Example

The effects on a Formula 1 car at a maximum speed of $\pm 360\ \mathrm{km/h}$





Measuring the race in two frames:

1) Spectator Frame (Earth)
2) Car's Frame

Assuming the car has a length of $\pm 5\ \mathrm{m}$

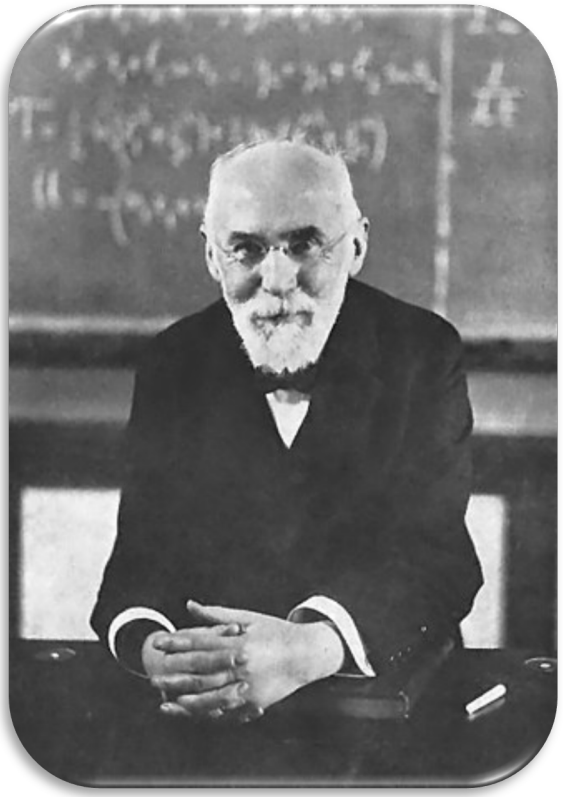
$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}} = \frac{1}{\sqrt{1 - \frac{100^2}{(3 \cdot 10^8)^2}}} \approx 1.0000000000000557$$

$$L = \frac{L_0}{\gamma} \approx \frac{5.0}{1.0000000000000557} \approx 4.999999999999722\ \mathrm{m}$$

Even at top F1 speed, relativity shortens the car by a distance far smaller than an atom!

Lorentz Transformations

Hendrik Antoon Lorentz
1853 - 1928



Lengths and times in a **moving frame of reference** depend on the **speed of the observer**. For an observer with speed u, the following equations apply:

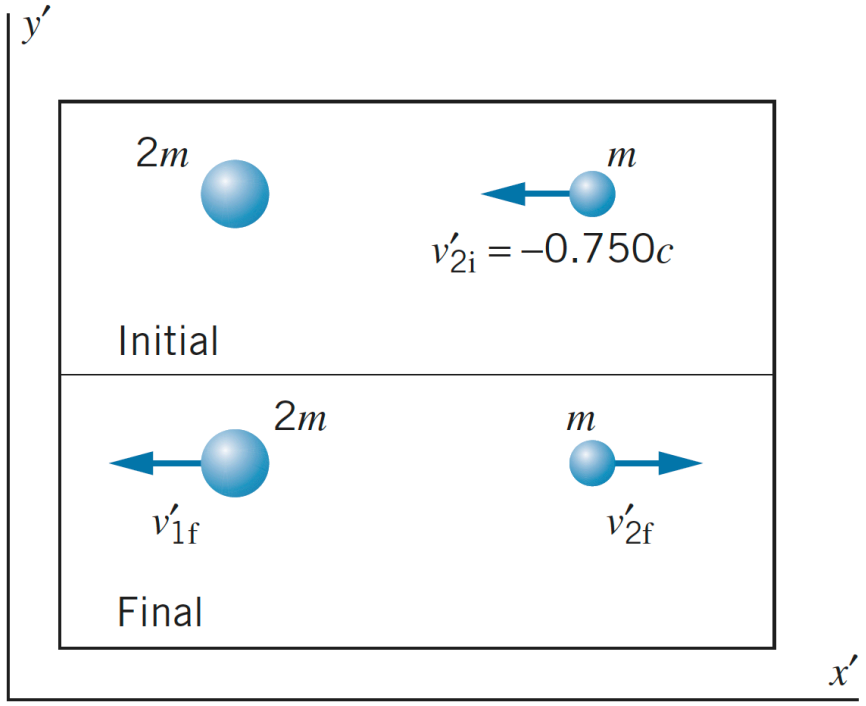
$$x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}} \qquad v_x' = \frac{v_x - u}{1 - v_x u/c^2}$$

$$y' = y \quad z' = z \qquad v_y' = \frac{v_y\sqrt{1 - u^2/c^2}}{1 - v_x u/c^2}$$

$$t' = \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}} \qquad v_z' = \frac{v_z\sqrt{1 - u^2/c^2}}{1 - v_x u/c^2}$$

Relativistic Dynamics

- Conservation of:
  - (Classical) Momentum: $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}$
  - (Classical) Kinetic Energy: $K = \frac{1}{2}mv^2$



By using: $m_1 v_{1i}^2 + m_2 v_{2i}^2 = m_1 v_{1f}^2 + m_2 v_{2f}^2$

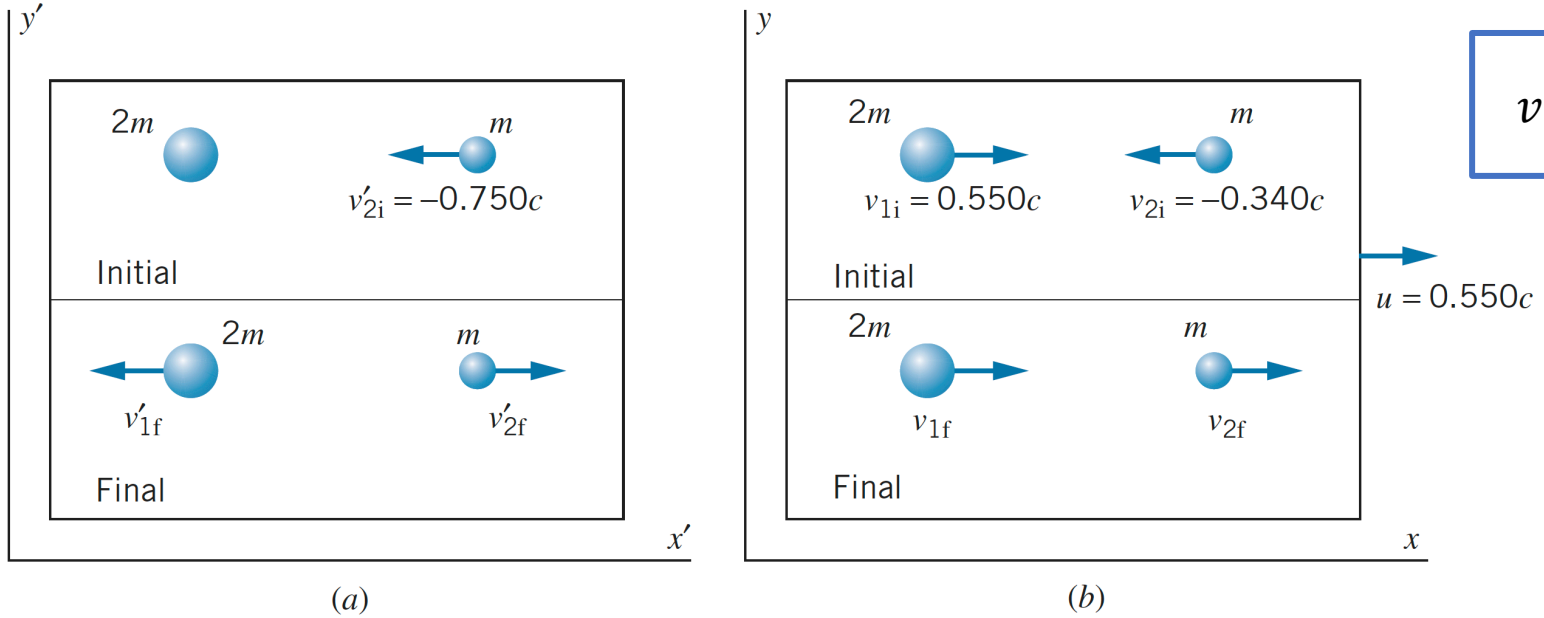
$$p'_i = m_1 v'_{1i} + m_2 v'_{2i} = (2m)(0) + (m)(-0.750c) = -0.750mc$$

$$p'_f = m_1 v'_{1f} + m_2 v'_{2f} = (2m)(-0.500c) + (m)(+0.250c) = -0.750mc$$

So $p'_i = p'_f$, check!

Relativistic Dynamics

- Now consider a moving frame of reference:



$$v = \frac{v' + u}{1 + v'u/c^2}$$

$$p_i = m_1 v_{1i} + m_2 v_{2i} = (2m)(+0.550c) + (m)(-0.340c) = +0.760mc$$

$$p_f = m_1 v_{1f} + m_2 v'_{2f} = (2m)(+0.069c) + (m)(+0.703c) = +0.841mc$$

$p_i \neq p_f$ → Violates Einstein's 1st Postulate!

Relativistic Dynamics

- We need to redefine momentum such that:
  1. Momentum conservation must be valid for all observers in inertial frames
  2. At low speeds the definition reduces to $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}$

- Same for the kinetic energy K

Relativistic Kinetic Energy

$$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

Relativistic Momentum

$$\vec{\boldsymbol{p}} = \frac{m\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}}$$

Relativistic Dynamics

- We can express $K = \frac{mc^2}{\sqrt{1-v^2/c^2}} - mc^2$ as $K = E - E_0$ with:
  - $E = \frac{mc^2}{\sqrt{1-v^2/c^2}}$ → Relativistic Total Energy
  - $E_0 = mc^2$ →Rest Energy

$$E = \sqrt{(pc)^2 + (mc^2)^2}$$

Particles without mass

$$E = pc$$

Massless particles have momentum and always travel at light speed!

Particles with mass

$$E = \sqrt{(pc)^2 + (E_0)^2}$$

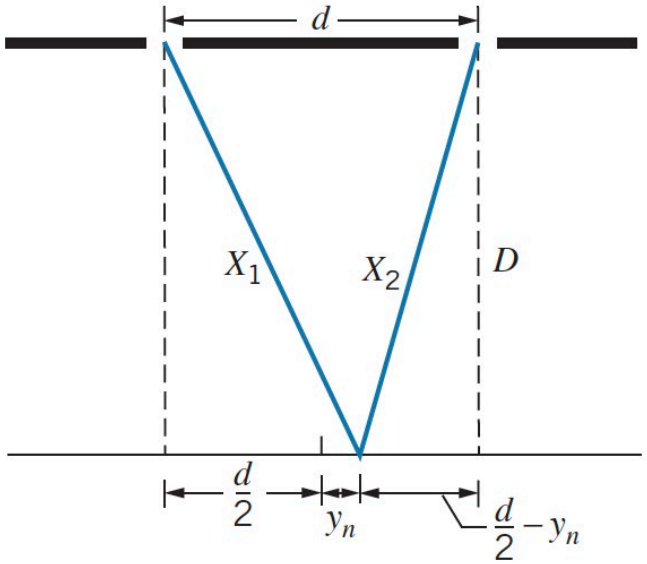
The Particle-like Properties of Electromagnetic Radiation

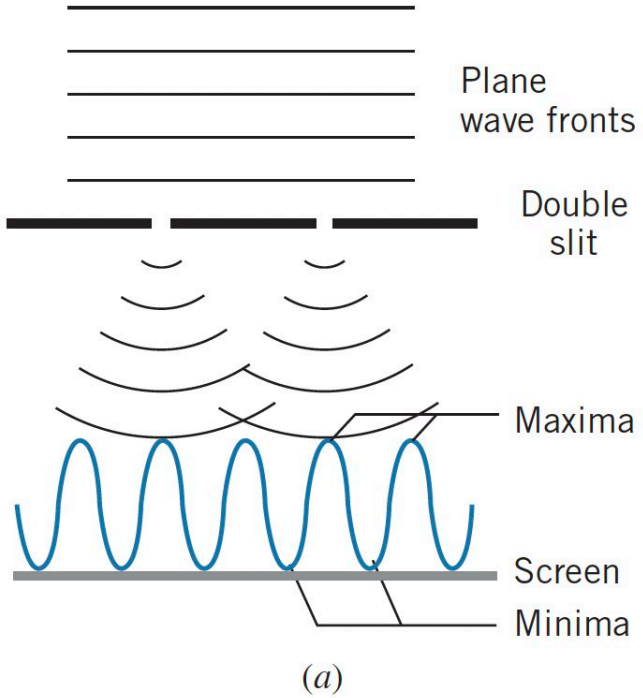
Light as Electromagnetic Wave

- Wave properties of interference and diffraction are shown in various experiments

- Young's double slit experiment

$$y_n = n\frac{\lambda D}{d}$$





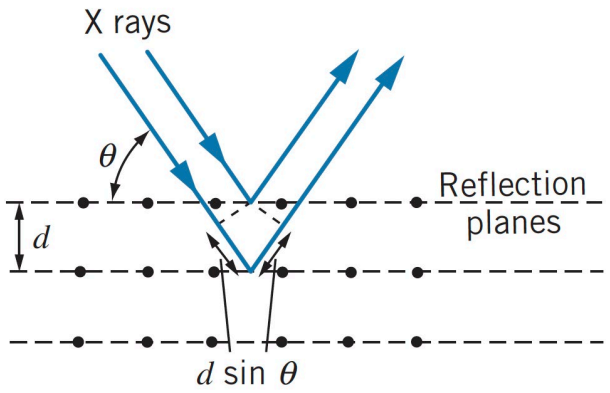
(a)

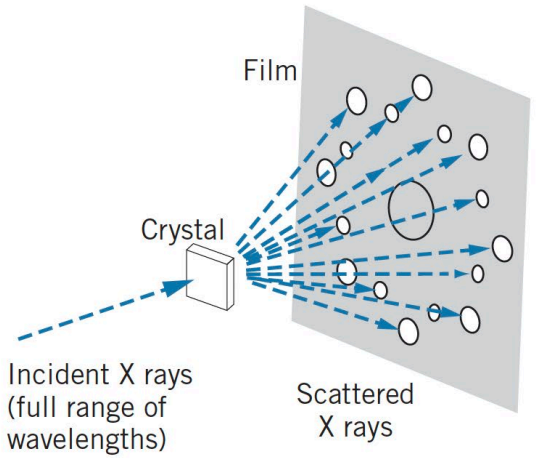


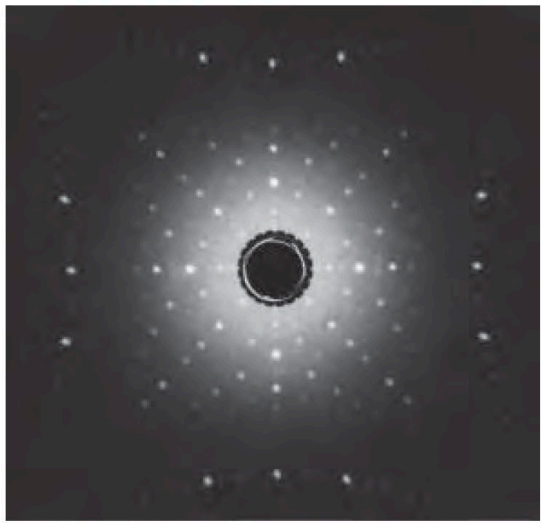
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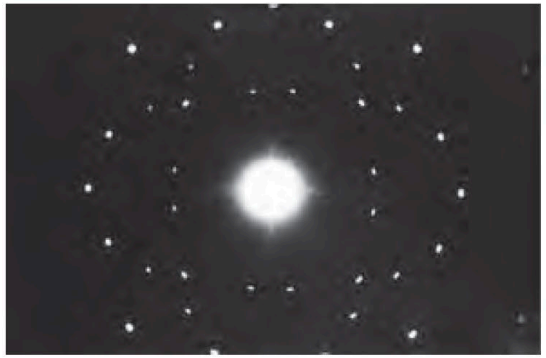
Light as Electromagnetic Wave

- A more modern version: atom lattice as ‘slits’



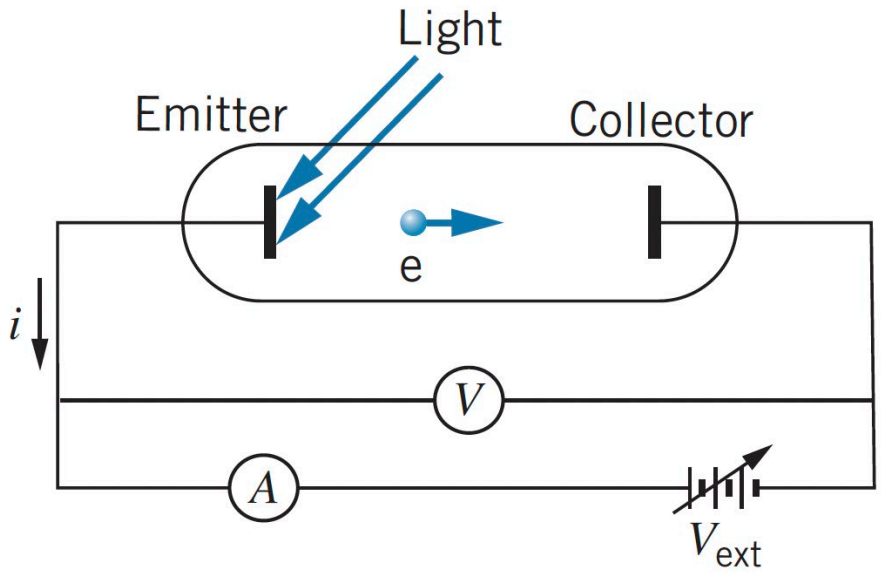






Photoelectric Effect

- One phenomenon that cannot be explained by the wave theory of light is the *Photoelectric Effect*
- Rate of electron emission measured by i
- Maximum kinetic energy measured by V
  - Stopping potential: $K_{max} = eV_s$



- Minimum quantity of energy needed to release an electron
  - Work function ϕ

Photoelectric Effect

- Work function is different for different materials:

Material	ϕ (eV)
Na	2.28
Al	4.08
Co	3.90
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14

Photoelectric Effect

- The photoelectric effect explained by wave theory:

1. *The maximum kinetic energy of the electrons should be proportional to the intensity of the radiation. (V_s depends on light intensity)*
2. *The photoelectric effect should occur for light of any frequency or wavelength. (similar effect for light of various frequencies)*
3. *The first electrons should be emitted in a time interval of the order of seconds after the radiation begins to strike the surface.*

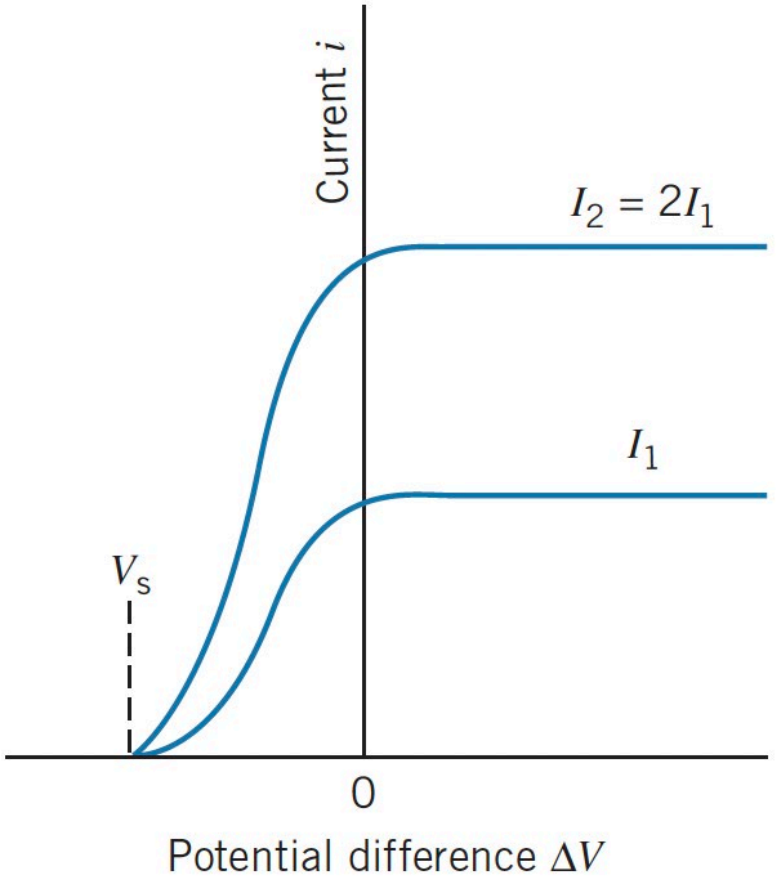
All of these theories do not comply with experimental results!

Photoelectric Effect

1. Doubling the intensity does not change the stopping potential (V_s)

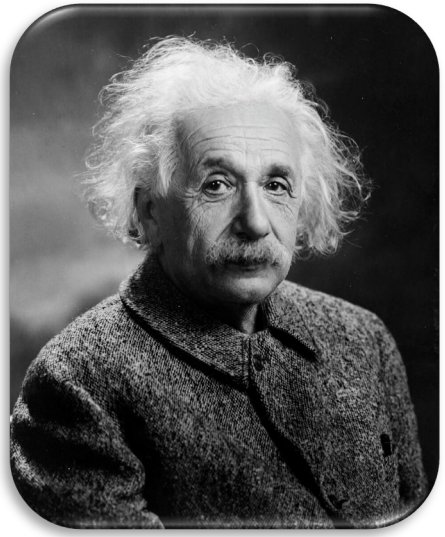
2. The photoelectric effect does not occur at all if the frequency of the light source is below a certain value
   - There is a so-called cutoff frequency f_c

3. The first photoelectrons are emitted instantaneously after the light source is turned on



Photoelectric Effect

Albert Einstein
1879 - 1955



- The photoelectric effect could not be explained with classical physics. Einstein came with an answer in 1905:

Light consists of small energy packages or *photons* with a fixed amount of energy

$$E = hf = h\frac{c}{\lambda} \quad \text{and} \quad p = \frac{E}{c} = \frac{h}{\lambda}$$

With $h = 6{,}62608 \cdot 10^{-34} Js$, Planck's constant

Photoelectric Effect – Summary

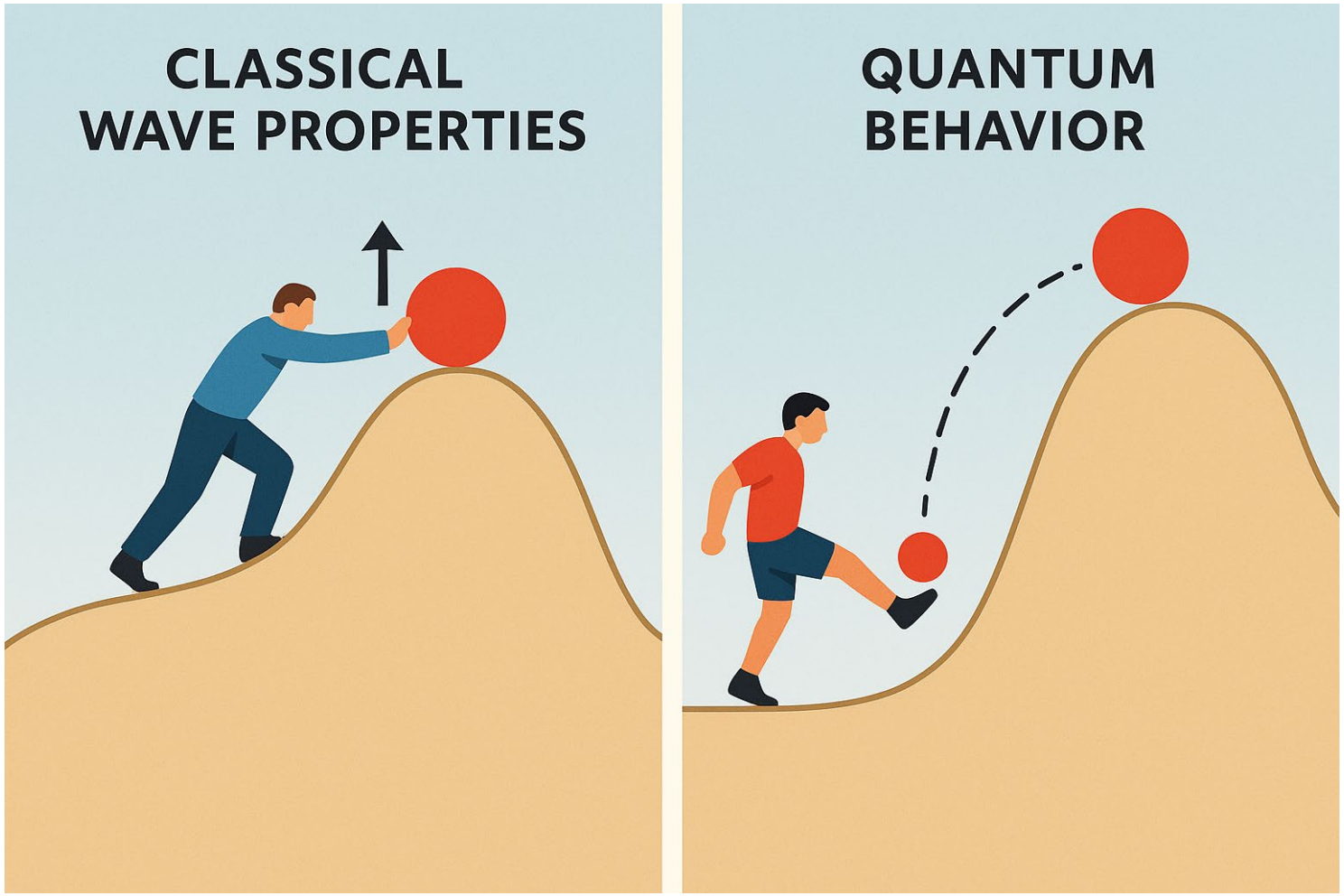
- A photon can be absorbed by an electron at the surface of a material. The electron can escape if the energy of the photon is larger than the energy needed to release the electron from the material (work function):

$$E_{photon} = hf > \phi$$

- Surplus energy is spent on various processes. In the border case, surplus energy is entirely transformed into kinetic energy, which is exactly enough to make it to the anode:

$$K_{max} = eV_s = hf - \phi$$

Photoelectric Effect – Summary



Photoelectric Effect – Exercise

A metal surface has a photoelectric cutoff wavelength of 325.6 nm. It is illuminated with light of wavelength 259.8 nm. What is the stopping potential?

$$K_{max} = eV_s = hf - \phi$$

Photoelectric Effect – Exercise

A metal surface has a photoelectric cutoff wavelength of 325.6 nm. It is illuminated with light of wavelength 259.8 nm. What is the stopping potential?

$$K_{max} = eV_s = hf - \phi$$

At the cutoff wavelength: $K_{max} = 0 \implies hf = \frac{hc}{\lambda_c} = \phi$

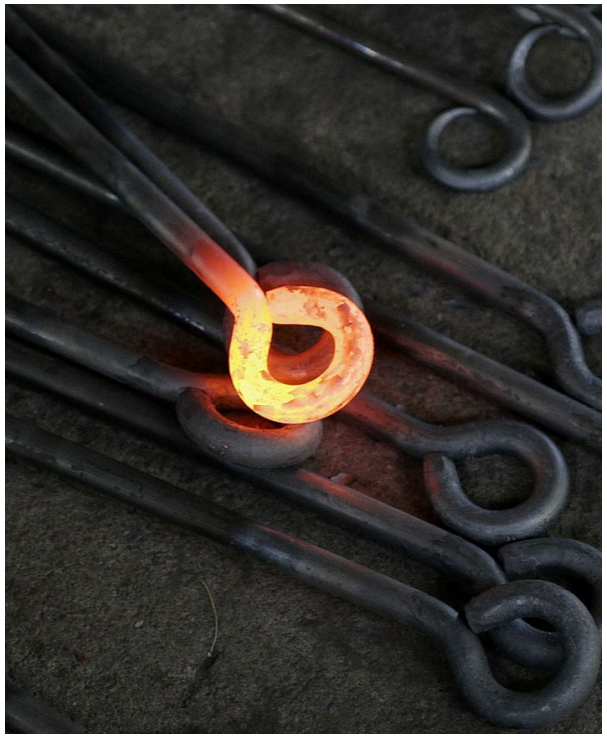
$$\phi = \frac{hc}{\lambda_c} = \frac{1240\ \text{eV} \cdot \text{nm}}{325.6\ \text{nm}} = 3.808\ \text{eV}$$

$$eV_s = \frac{hc}{\lambda} - \phi = \frac{1240\ \text{eV}\cdot\text{nm}}{259.8\ \text{nm}} - 3.808\ eV = 0.965\ \text{eV}$$

$$V_s = 0.965\ \text{V}$$

Thermal Radiation

- A second experiment that cannot be explained by classical wave theory: *thermal radiation*



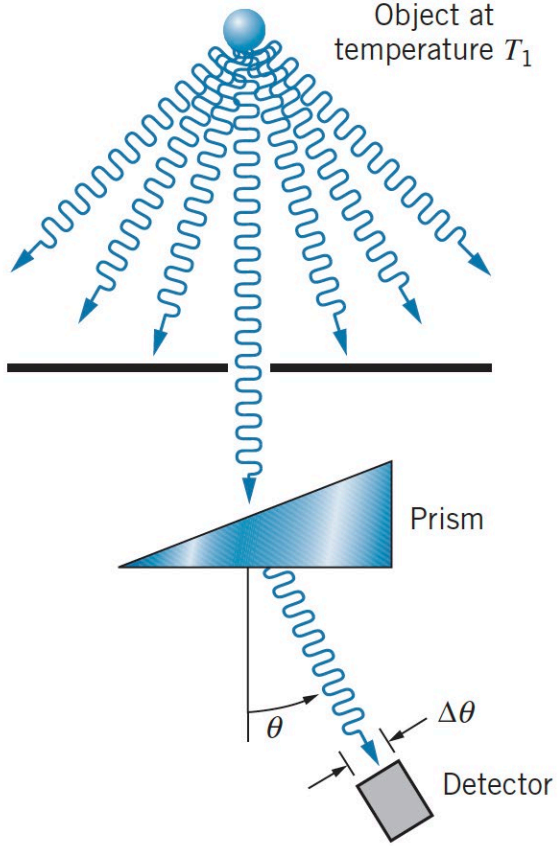


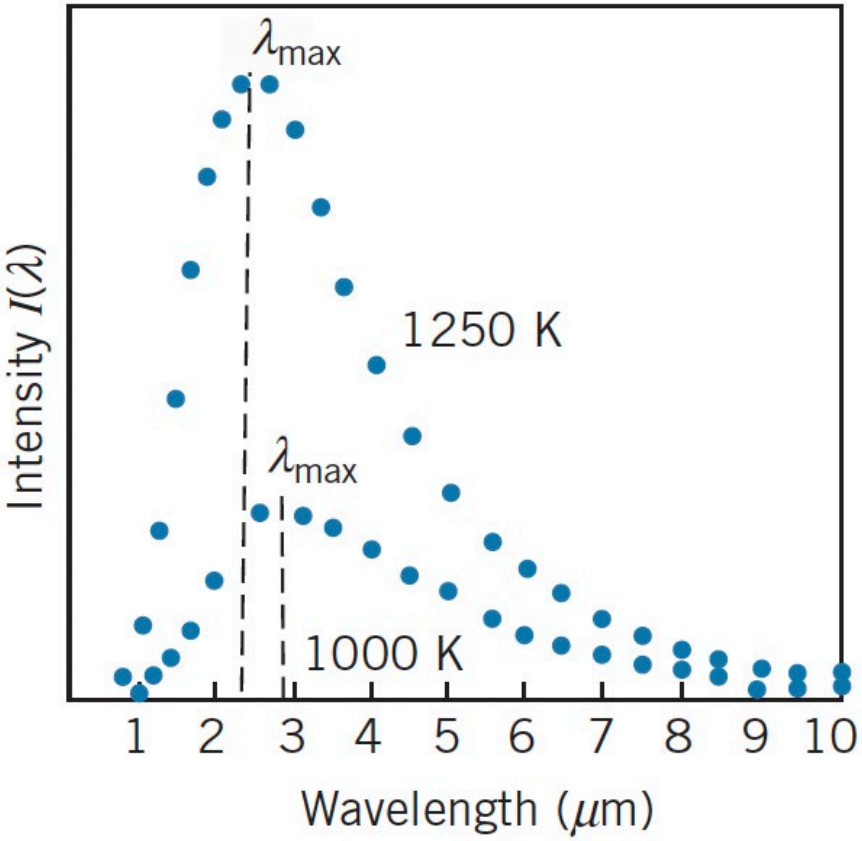
FIGURE 3.12 Measurement of the spectrum of thermal radiation. A device such as a prism is used to separate the wavelengths emitted by the object.

Thermal Radiation

The results show two important characteristics:

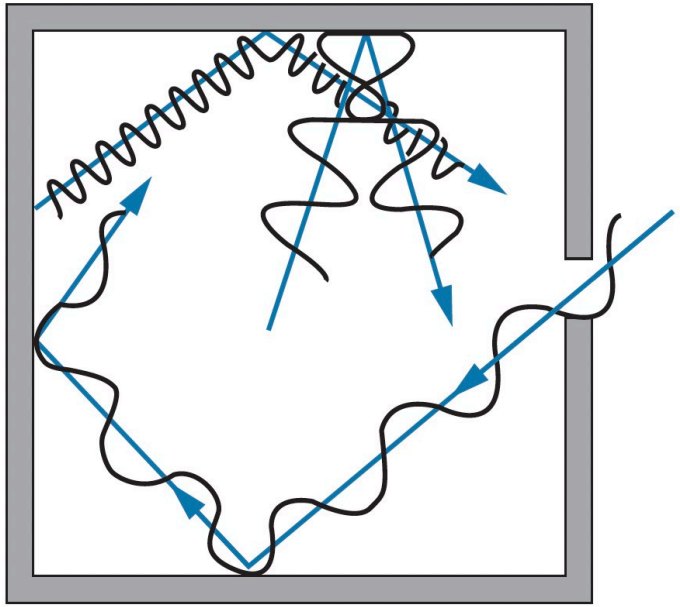
1. The total intensity radiated over all wavelengths (area under the curve) increases with increasing temperature according to Stefan's Law:
   $I = \sigma T^4$, with $\sigma = 5.67037 \cdot 10^{-8} \frac{W}{m^2} \cdot K^4$

2. The wavelength λ_{max} decreases as the temperature is increased according to Wien's discplacement law:
   $\lambda_{max} T = 2.8978 \cdot 10^{-3}\ m \cdot K$



Thermal Radiation – Classical Approach

- Thermal radiation of an arbitrary object is extremely difficult
- Simplify the analysis by defining a *blackbody*
- Following classical theory, we can find the corresponding intensity per unit wavelength:



Rayleigh-Jeans formula

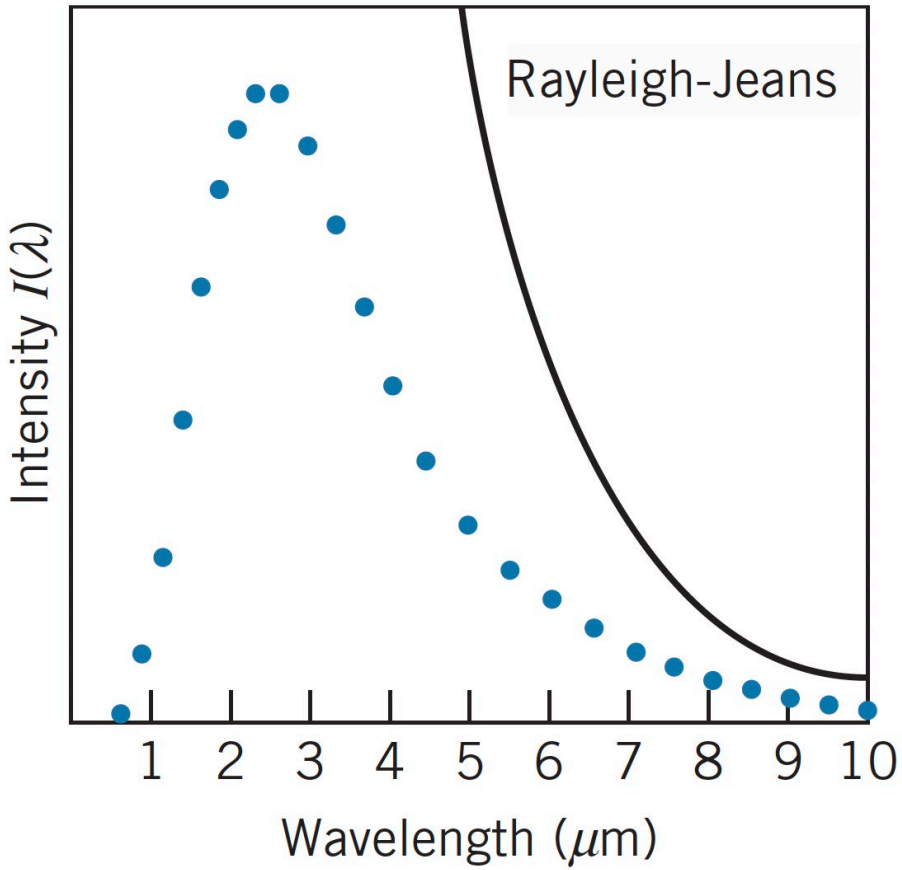
$$I(\lambda) = \frac{2\pi c}{\lambda^4} kT$$

(see chapter 3.3 for derivation)

Thermal Radiation – Classical Approach

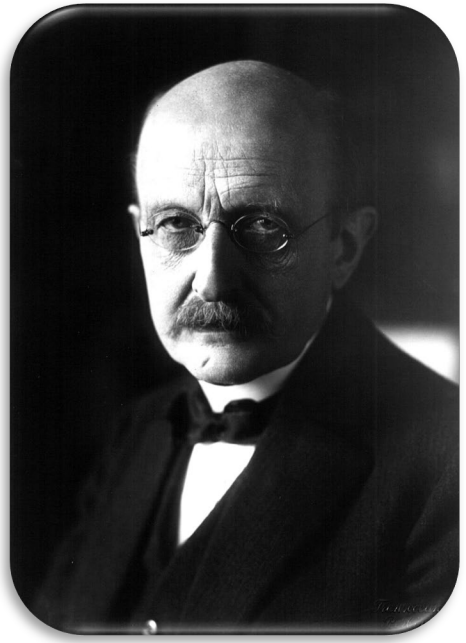
$$I(\lambda) = \frac{2\pi c}{\lambda^4} kT$$

Only accurate for long wavelengths!



Thermal Radiation – Quantum Approach

Max Planck
1858-1947



- Max Planck found a solution by making the following assumptions:
  1. An oscillator (atom) can only absorb energy in discrete amounts, or *quanta*. Energy is absorbed/radiated in fixed amounts $E = n\,\varepsilon$ with n =1, 2, 3,…
  2. The energy quanta are frequency-dependent: $\varepsilon = h\,f$

- Planck's formula fit perfectly with experimental data:

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT} - 1}$$

Thermal Radiation – Quantum Approach

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT} - 1}$$

- Deducing Stefan's Law from Planck's formula results in:

$$\sigma = \frac{2\pi^5 k^4}{15c^2h^3}$$

- Planck could determine the value for h from experiments:

$$h = 6.56 \cdot 10^{-34} J \cdot s$$

- Same as for the photoelectric effect experiment!

